

Exercises and Solutions

Theory of Distributions

Prepared by ...

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Exercise 1.1

Consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos[(\pi-\theta)z]}{2\sin(\pi z)}.$$

a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$. At $z = \pm i$:

$$\text{Res}[F, \pm i] = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

At $z = n \in \mathbb{Z}$:

$$\text{Res}[F, n] = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

b) The contour $\Gamma^{(N)}$

Define

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad -1 \leq t \leq 1,$$

$$\Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i), \quad -1 \leq t \leq 1,$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad -1 \leq t \leq 1,$$

$$\Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i), \quad -1 \leq t \leq 1.$$

Then $\Gamma^{(N)}$ is the closed square contour with vertices $\pm(N + \frac{1}{2}) \pm i(N + \frac{1}{2})$. It encloses the poles at $\pm i$ and all integers n with $|n| \leq N$.

c) Bound on the contour integral

On $\Gamma^{(N)}$, $|z| \sim N$, so $|F(z)| = O(1/N^2)$. Since the contour has length $O(N)$, we obtain

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N},$$

for some constant C independent of N .

d) Residue theorem and Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma^{(N)}} F(z) dz = 2\pi i \left(\text{Res}[F, i] + \text{Res}[F, -i] + \sum_{n=-N}^N \text{Res}[F, n] \right).$$

Hence

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{1 + n^2} \right| \leq \frac{C}{N}.$$

Letting $N \rightarrow \infty$, we conclude

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2}.$$

This gives the Fourier series expansion, uniformly in θ .

Exercise 1.1

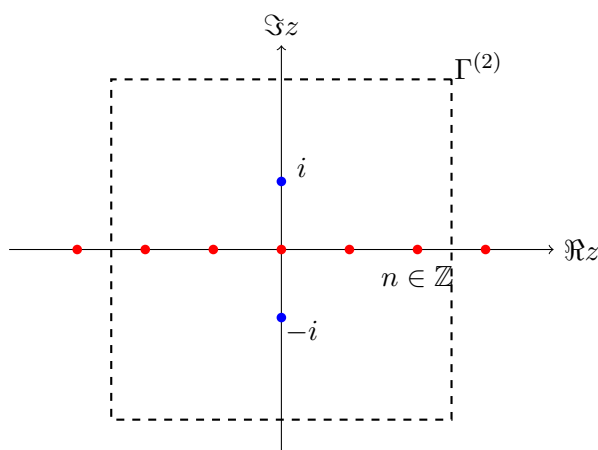
Consider the meromorphic function

$$F(z) = \frac{1}{1 + z^2} \frac{\cos[(\pi - \theta)z]}{2 \sin(\pi z)}.$$

a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$

Chart: Poles and Contour



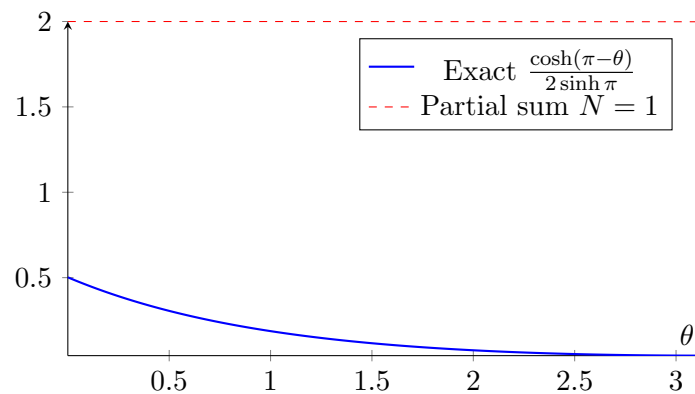
Legend. Red dots = poles at integers, Blue dots = poles at $\pm i$, dashed square = contour $\Gamma^{(N)}$.

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d) Residue theorem and Fourier series

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Chart: Fourier Series vs. Exact Expression

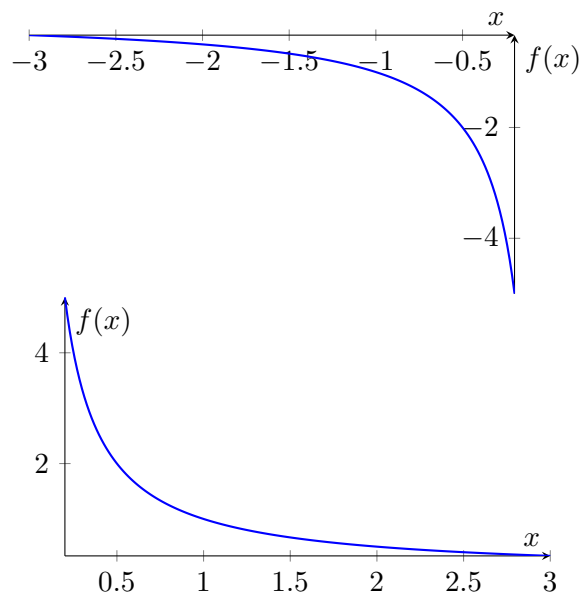


Legend. Blue curve = exact hyperbolic expression, red dashed = Fourier sum with $N = 1$, green dashed = Fourier sum with $N = 5$.

Introductory Charts: Simple Meromorphic Functions

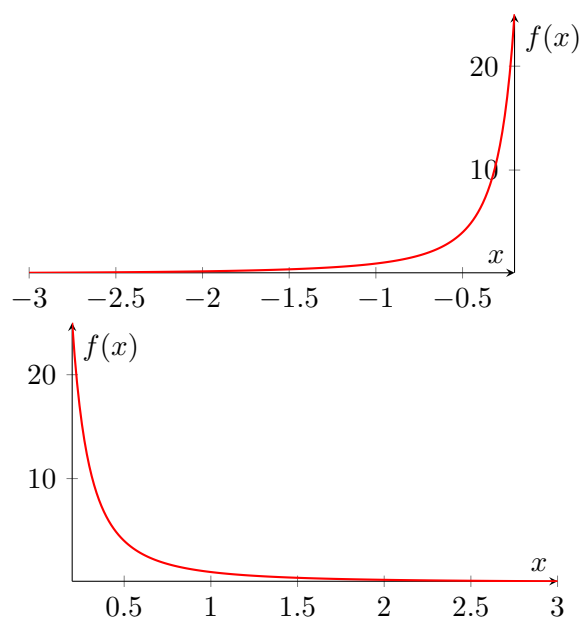
To recall the behavior of basic meromorphic functions near poles, we show plots of $f(z) = 1/z$ and $f(z) = 1/z^2$ on the real line. Both have a pole at $z = 0$, with $1/z$ being of order 1 and $1/z^2$ being of order 2. Clearly, $(1/z)^2 = 1/z^2$.

Chart: $f(x) = 1/x$ on $\mathbb{R} \setminus \{0\}$

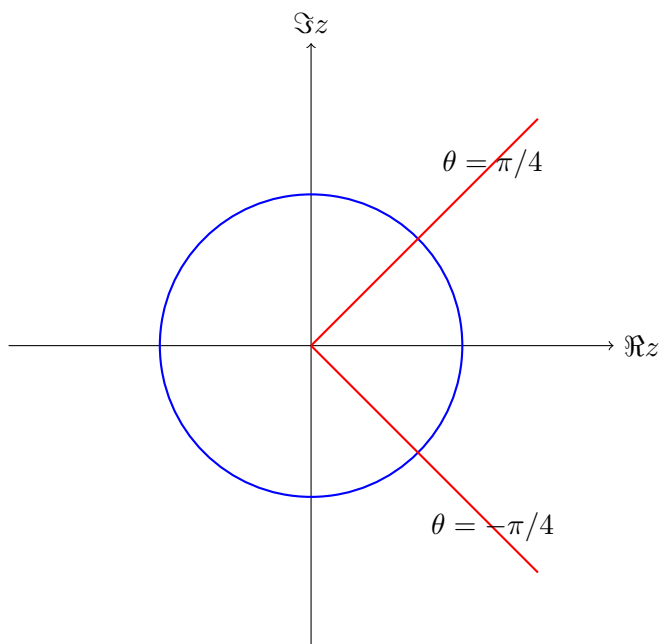


Legend. Function $1/x$ has a simple pole at $x = 0$. The left and right limits blow up to $\mp\infty$.

Chart: $f(x) = 1/x^2$



Legend. Function $1/x^2$ has a pole of order 2 at $x = 0$. It diverges to $+\infty$ on both sides of the pole. Clearly, $(1/x)^2 = 1/x^2$.



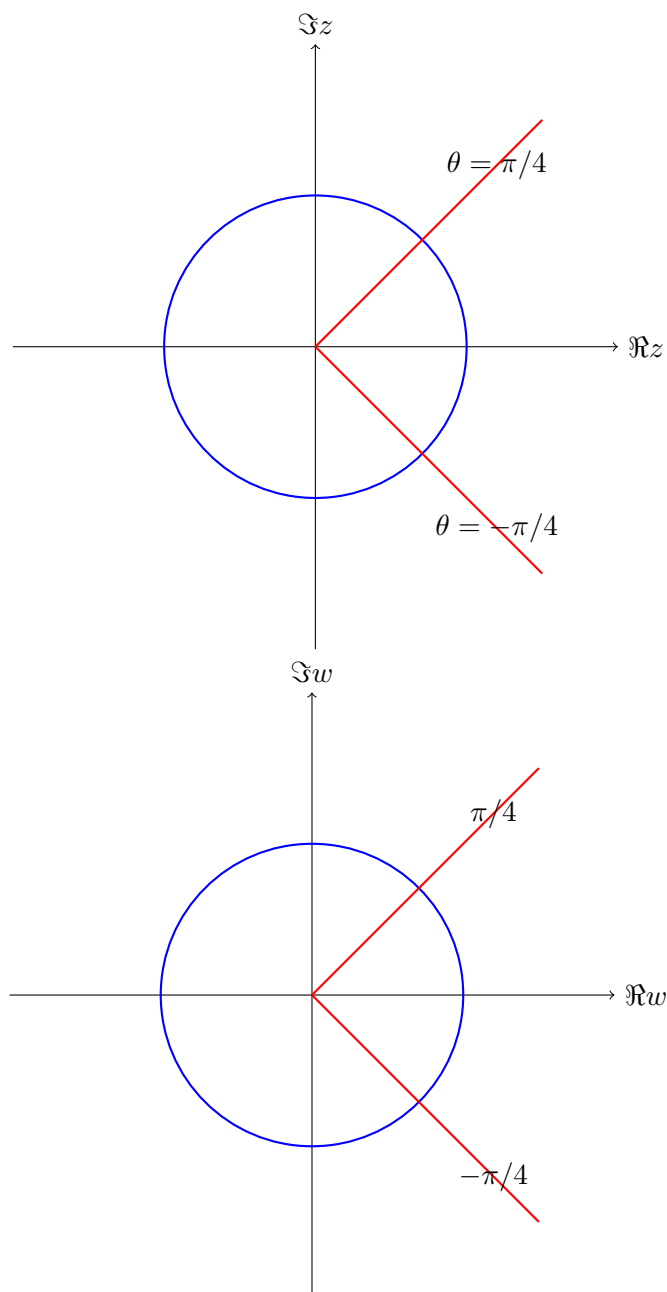
Introductory Examples: Simple Meromorphic Maps

Example 1: $f(z) = \frac{1}{z}$

Description. For $z = re^{i\theta}$ we have

$$\frac{1}{z} = \frac{1}{r}e^{-i\theta}.$$

Thus $f(z)$ inverts the magnitude ($r \mapsto 1/r$) and reflects the angle ($\theta \mapsto -\theta$). The unit circle maps to itself with reversed orientation. Lines through the origin are mapped to lines through the origin, reflected in the real axis.



Legend. Left: domain (z -plane). Right: image ($w = 1/z$). The unit circle is preserved, rays reflect across the real axis.

Example: $f(z) = \frac{1}{z}$

Description. For $z = re^{i\theta}$ we have

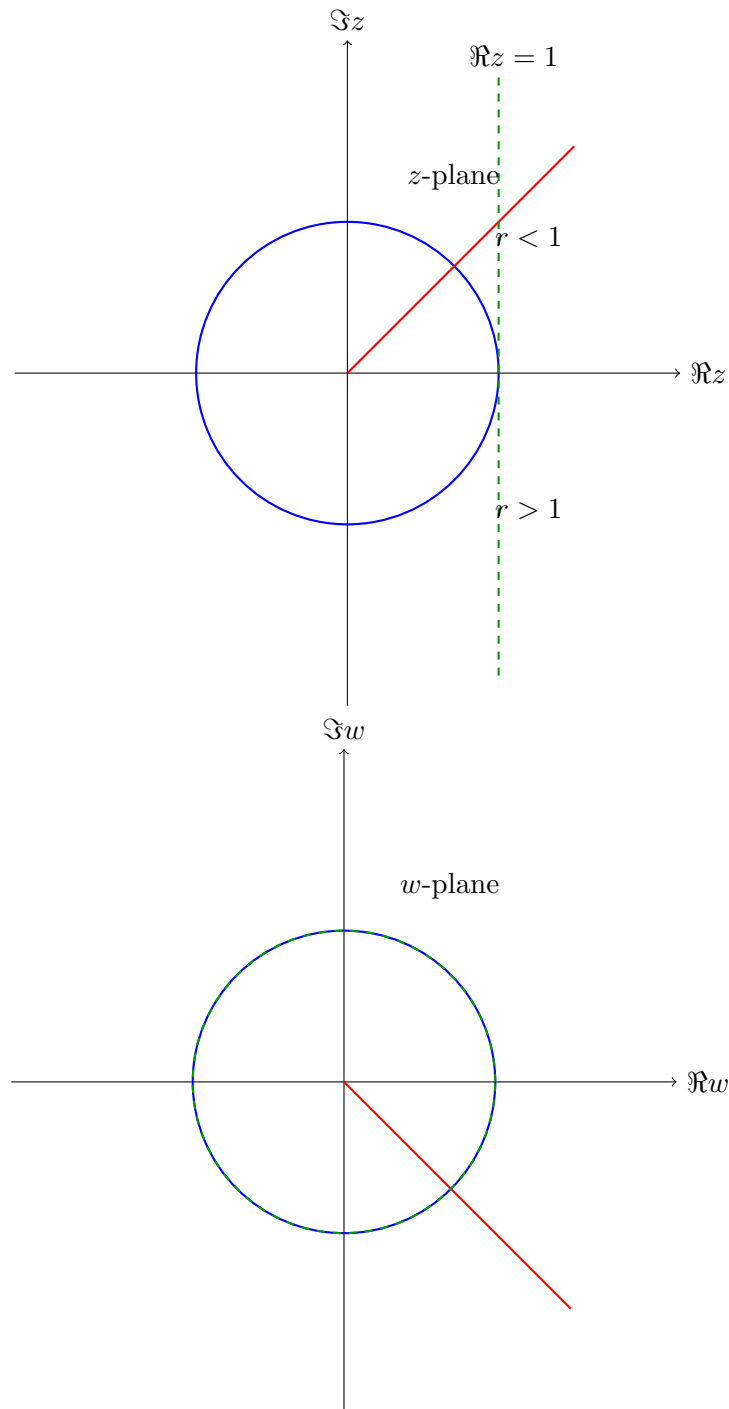
$$\frac{1}{z} = \frac{1}{r}e^{-i\theta}.$$

Thus the mapping $f(z) = 1/z$ has the following geometric effects:

- **Inversion of radii:** $r \mapsto 1/r$.
- **Reflection of angles:** $\theta \mapsto -\theta$.
- The unit circle $|z| = 1$ is invariant, but orientation is reversed.
- The interior of the unit disk $|z| < 1$ maps to the exterior $|w| > 1$, and vice versa.

- Rays through the origin map to reflected rays across the real axis.
- More generally, lines not through the origin map to circles, and circles not through the origin map to circles (inversion geometry).

Chart: Domain vs. Image under $1/z$



Legend. Blue: unit circle (invariant). Red: ray at $\theta = \pi/4$ mapping to $\theta = -\pi/4$. Green dashed: vertical line $\Re z = 1$ maps to a circle through the origin.

Example: $f(z) = \frac{1}{z+1}$

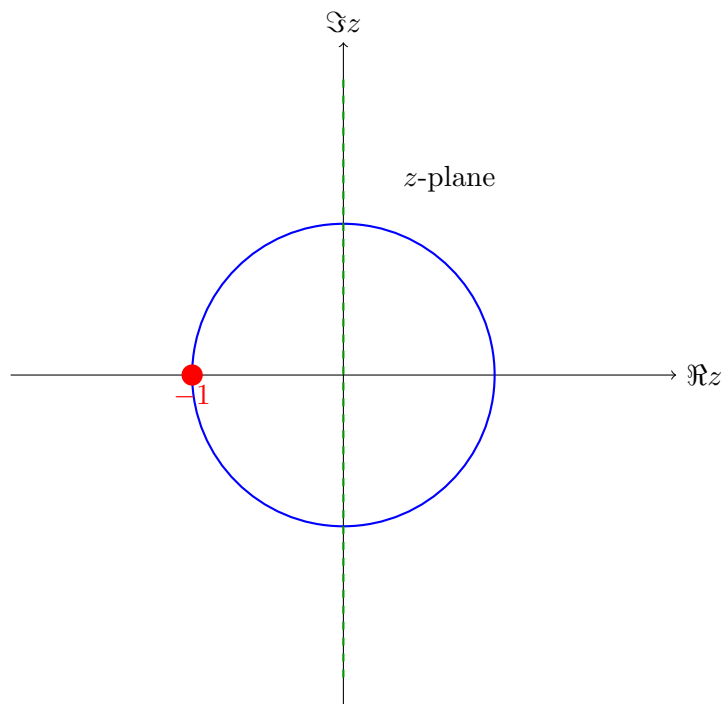
Description. This mapping is a shifted inversion. Let $w = z + 1$. Then

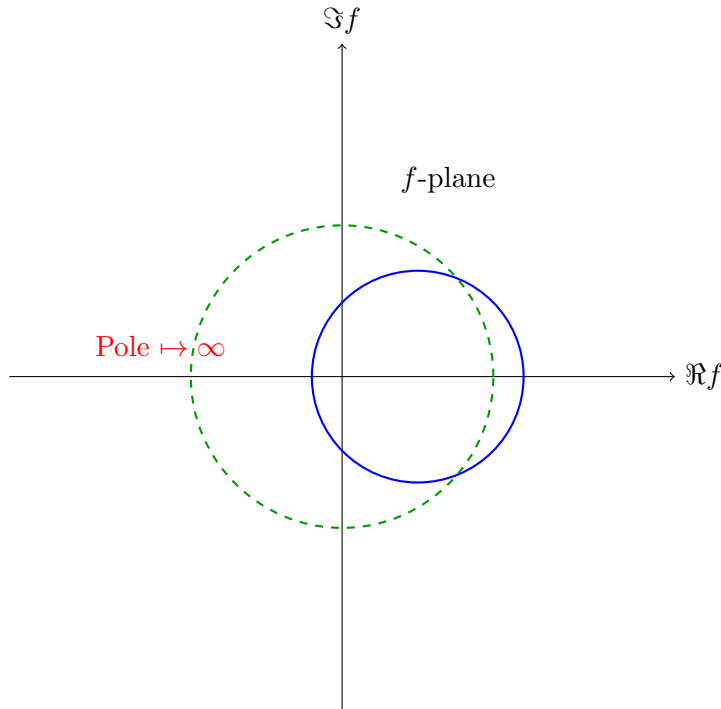
$$f(z) = \frac{1}{z+1} = \frac{1}{w}.$$

Thus the inversion is centered at $z = -1$ instead of $z = 0$. The function has a simple pole at $z = -1$. As $|z| \rightarrow \infty$, $f(z) \rightarrow 0$. Key geometric effects:

- Pole at $z = -1$ corresponds to $w = 0$ under translation.
- The unit circle $|z| = 1$ no longer maps to itself, but to another circle in the w -plane.
- Lines and circles not passing through $z = -1$ map to circles.
- Rays and bands relative to $z = -1$ are inverted accordingly.

Chart: Domain vs. Image under $1/(z+1)$





Legend. Left: domain (z -plane) with unit circle (blue), pole at $z = -1$ (red), and imaginary axis (green dashed). Right: schematic image under $f(z) = 1/(z + 1)$: unit circle maps to another circle, imaginary axis maps to a circle through the origin, and $z = -1$ maps to infinity.

Example: $f(z) = \frac{1}{z^2}$

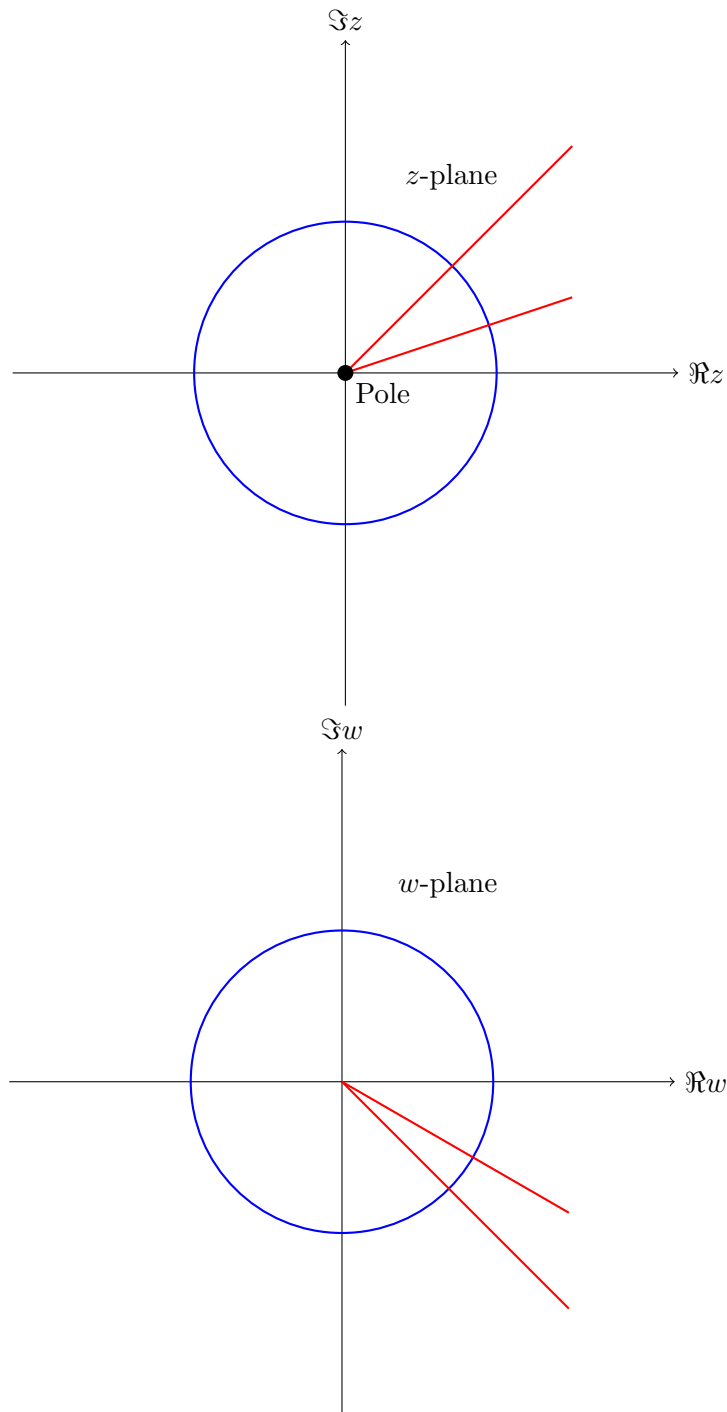
Description. For $z = re^{i\theta}$ we have

$$f(z) = \frac{1}{z^2} = \frac{1}{r^2} e^{-2i\theta}.$$

Thus the mapping $f(z) = 1/z^2$ has the following geometric effects:

- **Inversion of radii (squared):** $r \mapsto 1/r^2$.
- **Angle doubling and reflection:** $\theta \mapsto -2\theta$.
- The unit circle $|z| = 1$ is invariant (maps to itself).
- The interior of the unit disk $|z| < 1$ maps to the exterior $|w| > 1$, and vice versa.
- Rays at angle θ map to rays at angle -2θ .
- There is a pole of order 2 at $z = 0$.

Chart: Domain vs. Image under $1/z^2$



Legend. Blue: unit circle (invariant). Red: sample rays mapped under $\theta \mapsto -2\theta$. Black dot: pole of order 2 at $z = 0$.

Exercise 1.2 (Periodic Green's function for $-\psi'' + \psi = f$)

Suppose $f \in C^0(\mathbb{R})$ is 2π -periodic, i.e. $f(\theta) = f(\theta + 2\pi)$. For $\theta \in [0, 2\pi)$ define

$$\psi(\theta) := \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha, \quad (1)$$

and extend ψ to $\mathbb{R}$ by periodicity $\psi(\theta + 2\pi) = \psi(\theta)$.

Background / definition. It is convenient to write (1) using the periodic Green kernel

$$G(\theta, \alpha) := \frac{\cosh(\pi - |\theta - \alpha|)}{2 \sinh \pi}, \quad \theta, \alpha \in [0, 2\pi). \quad (2)$$

Then (1) is equivalently

$$\psi(\theta) = \int_0^{2\pi} f(\alpha) G(\theta, \alpha) d\alpha, \quad (3)$$

i.e. $\psi = L^{-1}f$ for the periodic operator $L := -\frac{d^2}{d\theta^2} + 1$.

Solution. **(a) $\psi \in C^0(\mathbb{R})$.** On $[0, 2\pi)$ the map $(\theta, \alpha) \mapsto f(\alpha) G(\theta, \alpha)$ is continuous and bounded. By dominated convergence, $\theta \mapsto \psi(\theta)$ is continuous. The periodic extension is continuous since both f and G are 2π -periodic in each variable.

(b) First derivative and $\psi \in C^1(\mathbb{R})$. Differentiate (1) by Leibniz' rule. Using $\partial_\theta \cosh(\pi - \theta + \alpha) = -\sinh(\pi - \theta + \alpha)$ and $\partial_\theta \cosh(-\pi - \theta + \alpha) = -\sinh(-\pi - \theta + \alpha)$, and noting that the boundary terms cancel because $\frac{\cosh \pi}{2 \sinh \pi} - \frac{\cosh \pi}{2 \sinh \pi} = 0$, we obtain

$$\psi'(\theta) = - \int_0^\theta f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha - \int_\theta^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha. \quad (4)$$

Again by dominated convergence, the right-hand side is continuous, hence $\psi \in C^1(\mathbb{R})$.

(c) Second derivative and the equation $-\psi'' + \psi = f$. Differentiate (4) once more. Using $\partial_\theta \sinh(\pi - \theta + \alpha) = -\cosh(\pi - \theta + \alpha)$ and $\partial_\theta \sinh(-\pi - \theta + \alpha) = -\cosh(-\pi - \theta + \alpha)$, Leibniz' rule yields

$$\begin{aligned} \psi''(\theta) &= -\frac{f(\theta) \sinh \pi}{2 \sinh \pi} + \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha \\ &\quad - \frac{f(\theta) \sinh(-\pi)}{2 \sinh \pi} + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha \\ &= -f(\theta) + \psi(\theta). \end{aligned}$$

Thus for all $\theta \in \mathbb{R}$, we have

$$-\psi''(\theta) + \psi(\theta) = f(\theta).$$

Since f is continuous, the display shows $\psi \in C^2(\mathbb{R})$.